

PRST Class (27/10/2020)

Hyper-geometric distribution.

Suppose there are N individuals of which N_1 individuals have certain characteristic.

Suppose we draw a sample of size n at random without replacement.
(WOR)

$(10, 3)$

$\begin{pmatrix} P_1 \\ \vdots \\ P_{10} \end{pmatrix} \quad (P_3, P_2, P_7)$

10 persons,
Sample size 3 WOR

$P_1, P_2, (P_3), P_4, P_5, P_6, P_7, P_8, P_9, P_{10}$

$P_1, (P_2), P_4, P_5, P_6, P_7, P_8, P_9, P_{10}$

$P_1, P_4, P_5, P_6, (P_7), P_8, P_9, P_{10}$

(P_3, P_2, P_7)

Success
O blood group
Failure
other bg except O

With replacement

$P_1, P_2, (P_3), P_4, \dots, P_{10}$

$P_1, (P_2), P_3, P_4, \dots, P_{10}$

$P_1, P_2, (P_3), P_4, \dots, P_{10}$

P_3

P_2

P_3

all names the ... 1

X : # individuals will possess the characteristic p in the sample of size m .

$N = 10$ | $Np = 6$ - 0 by
4 - other by
" $N(1-p)$

N $\frac{Np}{n}$ $\frac{Nq}{m-n}$

(m)

$m = 3$

$$P(X=n) = \frac{\binom{Np}{n} \binom{Nq}{m-n}}{\binom{N}{m}} \quad | \quad n = 0, 1, 2, 3$$

$$\frac{Np \times (Np-1) \times (Np-2) \times \dots \times (Np-(n-1))}{n!} = \binom{Np}{n}$$

10 people
 $P(X=2)$

$6 \times 5 \times 4$

6 by 4 other by
 $m = 3$

0 $p_1 p_2 p_3 p_4 p_5 p_6$

o.w $p_7 p_8 p_9 p_{10}$

$$\frac{(p_1, p_3) + (p_3, p_1)}{2!}$$

$$\binom{N}{m} = \frac{N(N-1)(N-2) \dots (N-(m-1))}{m!}$$

$(p_1, p_2) (p_2, p_1)$

(p_1, p_2, p_3)
 (p_1, p_3, p_2) 6 ways
 (p_2, p_1, p_3)

$$(p_1, p_2) \quad (p_2, p_1)$$

$$\begin{pmatrix} p_1 & p_3 & p_2 & \dots & 1 \\ p_2 & p_1 & p_3 & & \\ \vdots & & & & \end{pmatrix}$$

$$E(X) = \sum_{x=0}^m x P(X=x)$$

$$(m < Np)$$

$$= \sum_{x=0}^m x \frac{\binom{Np}{x} \binom{Nq}{m-x}}{\binom{N}{m}}$$

10 per
6 - 0 by
4 - 0 by
m=8
x > Np

$$= m \times p$$

$$E(X(X-1))$$

$$\mu_2 = mpq \frac{N-m}{N-1}$$

Ex. Suppose your friend has 10 cookies and 7 of which are chocolate cookies. Your friend divides the cookies equally between you and him. What is the prob that you get all chocolate cookies?

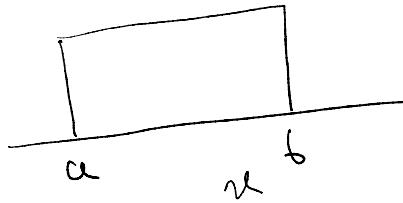
$$\text{ans} \rightarrow \frac{\binom{7}{5}}{\binom{10}{5}}$$

Continuous dist

Uniform dist

X : r.v. on $[a, b]$

$$f(x) = \frac{1}{b-a} ; a \leq x \leq b$$



$$F(x) = \int_a^x f(x) dx = \frac{x-a}{b-a}, \quad a \leq x \leq b$$